不定积分的积分法

换元法

第一类换元积分法

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \int \frac{1}{\sqrt{1-(\frac{x}{a})^2}} d(\frac{x}{a}) = \arcsin\frac{x}{a} + C$$

$$\int \frac{dx}{a^2+x^2} = \frac{1}{a}\int \frac{1}{1+(\frac{x}{a})^2} d(\frac{x}{a}) = \frac{1}{a}\arctan\frac{x}{a} + C$$

$$\int \frac{x+1}{x^2+2x+3}dx$$

$$\begin{aligned}\text{原式} =& \frac{1}{2}\int \frac{d(x^2+2x+3)}{x^2+2x+3} \\ =& \frac{1}{2}\ln(x^2+2x+3)+C\end{aligned}$$

$$\int \frac{x^2}{\sqrt{x^3+1}}dx$$

$$\begin{aligned}\text{原式} =& \frac{1}{3}\int \frac{d(x^3+1)}{\sqrt{x^3+1}} \\ =& \frac{2}{3}\int \frac{d(x^3+1)}{2\sqrt{x^3+1}} \\ =& \frac{2}{3}\sqrt{x^3+1}+C\end{aligned}$$

$$\int \frac{x}{4+x^4}dx$$

$$\begin{aligned}\text{原式} =& \frac{1}{2}\int \frac{d(x^2)}{2^2+(x^2)^2} \\ =& \frac{1}{4}\int \frac{d(x^2)}{2^2+(x^2)^2} \\ =& \frac{1}{4}\arctan\frac{x^2}{2}+C\end{aligned}$$

$$\int \frac{dx}{\sqrt{x}(1+x)}$$

$$\begin{aligned}\text{原式} =& 2\int \frac{dx}{2\sqrt{x}(1+x)} \\ =& 2\int \frac{d\sqrt{x}}{(1+x)} \\ =& 2\arctan\sqrt{x}+C\end{aligned}$$

$$\int \frac{\tan^2\sqrt{x}}{\sqrt{x}}dx$$

$$\begin{aligned}
\text{原式} =& 2\int \tan^2\sqrt{x}d\sqrt{x} \\
=& 2\int (\sec^2\sqrt{x}-1)d\sqrt{x} \\
=& 2(\tan\sqrt{x}-\sqrt{x})+C
\end{aligned}$$

$$\begin{aligned}
x^a &= (\frac{1}{a+1}x^{a+1})' \\
\frac{1}{2\sqrt{x}} &= (\sqrt{x})' \\
e^x &= (e^x)'
\end{aligned}$$

$$\int \frac{e^x}{4+e^{2x}}dx$$

$$\begin{aligned}
\text{原式} =& \int \frac{de^x}{2^2+(e^x)^2} \\
=& \frac{1}{2}\arctan\frac{e^x}{2}+C
\end{aligned}$$

$$\int \frac{1}{1+e^x}dx$$

$$\begin{aligned}
\text{原式} =& \int \frac{dx}{e^x(e^{-x}+1)} \\
=& \int \frac{e^{-x}}{e^{-x}+1}dx \\
=& -\int \frac{d(e^{-x}+1)}{e^{-x}+1} \\
=& -\ln(e^{-x}+1)+C
\end{aligned}$$

$$\text{计算不定积分} \int \frac{\sin x}{\sin x+\cos x}dx.$$

$$\begin{aligned}
\text{原式} =& \frac{1}{\sqrt{2}}\int \frac{\sin x}{\cos(x-\frac{\pi}{4})}dx \\
=& \frac{1}{\sqrt{2}}\int \frac{\sin[(x-\frac{\pi}{4})+\frac{\pi}{4}]}{\cos(x-\frac{\pi}{4})}d(x-\frac{\pi}{4}) \\
=& \frac{1}{2}\int \frac{\sin(x-\frac{\pi}{4})+\cos(x-\frac{\pi}{4})}{\cos(x-\frac{\pi}{4})}d(x-\frac{\pi}{4}) \\
=& \frac{1}{2}\int \tan(x-\frac{\pi}{4})d(x-\frac{\pi}{4})+\frac{1}{2}(x-\frac{\pi}{4}) \\
=& -\frac{1}{2}\ln|\cos(x-\frac{\pi}{4})|+\frac{x}{2}+C
\end{aligned}$$

$$F'(u)=f(u),u=\Phi(x)\text{可导}$$
$$\int f[\Phi(x)]\Phi'(x)dx=\int f[\Phi(x)]d\Phi(x)=\int f(t)dt$$
$$=F(t)+C=F[\Phi(x)]+C$$

$$\int\frac{dx}{x^2+2x+5}$$
$$\begin{aligned}\text{原式}=&\int\frac{d(x+1)}{2^2+(x+1)^2}\\=&\frac{1}{2}\arctan\frac{x+1}{2}+C\end{aligned}$$

$$\int\frac{dx}{x^2-x-2}$$
$$\begin{aligned}\text{原式}=&\int\frac{dx}{(x-2)(x+1)}\\=&\frac{1}{3}\int(\frac{1}{x-2}-\frac{1}{x+1})dx\\=&\frac{1}{3}[\int\frac{d(x-2)}{x-2}-\int\frac{d(x+1)}{x+1}]\\=&\frac{1}{3}\ln|\frac{x-2}{x+1}|+C\end{aligned}$$

$$\int\frac{dx}{\sqrt{x}(4+x)}$$
$$\begin{aligned}\text{原式}=&2\int\frac{d\sqrt{x}}{2^2+(\sqrt{x})^2}\\=&\arctan\frac{\sqrt{x}}{2}+C\end{aligned}$$

$$\int\frac{dx}{x\ln^2x}$$
$$\begin{aligned}\text{原式}=&\int\frac{d(\ln x)}{\ln^2x}\\=&-\frac{1}{\ln x}+C\end{aligned}$$

第二类换元积分法

$$\int f(x)dx=\int f[\Phi(t)]\Phi'(t)dt=\int g(t)dt$$
$$=G(t)+C=G[\Phi^{-1}(x)]+C$$

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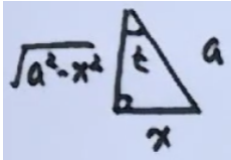
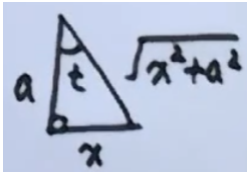
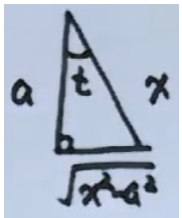
1. 无理 $\Rightarrow$ 有理

$$\int \frac{dx}{\sqrt{x(1-x)}} = 2\int \frac{dx}{2\sqrt{x}\sqrt{1-x}} = 2\frac{d(\sqrt{x})}{\sqrt{1-(\sqrt{x})^2}}$$
$$= 2\arcsin\sqrt{x} + C$$

$$求 \int \frac{dx}{1+\sqrt{x}}.$$
$$t = \sqrt{x}$$
$$\int \frac{dx}{1+\sqrt{x}} = 2\int \frac{tdt}{1+t} = 2\int (1-\frac{1}{1+t})dt = 2(t-\ln|1+t|) + C = 2\sqrt{x} - 2\ln(1+\sqrt{x}) + C$$





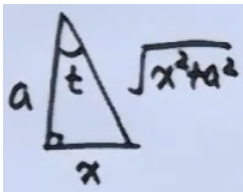


2. 三角代换

①$\sqrt{a^2-x^2} = a\cos t, x = a\sin t$

②$\sqrt{x^2+a^2} = a\sec t, x = a\tan t$

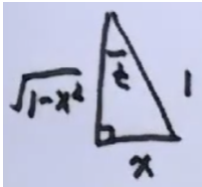
③$\sqrt{x^2-a^2} = a\tan t, x = a\sec t$



$$\int \frac{dx}{\sqrt{x^2+a^2}} = \int \frac{dx}{a\sec t}, x = a\tan t$$
$$= \int \frac{a\sec^2 t}{a\sec t}dx = \int \sec t dt = \ln|\sec t + \tan t| + C$$
$$= \ln\frac{\sqrt{x^2+a^2}+x}{a} + C$$
$$= \ln(x+\sqrt{x^2+a^2}) + C$$

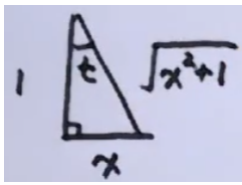
$$求不定积分 \int \frac{dx}{x^2\sqrt{1-x^2}}.$$

$$
\begin{aligned}
\text{原式} &= \int \frac{\cos t}{\sin^2 t \cos t} dt, x = \sin t \\
&= \int \csc^2 dt \\
&= -\cot t + C \\
&= -\frac{\sqrt{1-x^2}}{x} + C
\end{aligned}
$$



$$
\text{求} \int \frac{1}{x^2\sqrt{x^2+1}} dx
$$

$$
\begin{aligned}
\text{原式} &= \int \frac{\sec^2 t}{\tan^2 t \sec t} dt, x = \tan t \\
&= \int \frac{\cos t}{\sin^2 t} dt \\
&= \frac{d(\sin t)}{\sin^2 t} \\
&= -\frac{1}{\sin t} + C \\
&= -\frac{\sqrt{x^2+1}}{x} + C
\end{aligned}
$$



分部积分法

$$
\begin{aligned}
(uv)' &= u'v + uv' \\
uv &= \int u'vdx + \int uv'dx \\
&= \int vdu + \int udv \\
\int udv &= uv - \int vdu
\end{aligned}
$$

$$
\int \text{幂} * \text{指} dx
$$

$$
\text{求} \int x^2 e^x dx.
$$

$$
\begin{aligned}
\int x^2e^xdx &= \int x^2d(e^x)\\
&=x^2e^x-2\int xe^xdx\\
&=x^2e^x-2\int xd(e^x)\\
&=x^2e^x-2(xe^x-\int e^xdx)\\
&=x^2e^x-2xe^x+2e^x+C
\end{aligned}
$$

$$\int \text{幂} * \text{对数} dx$$

$$\text{求} \int x\ln^2 xdx.$$

$$
\begin{aligned}
\int x\ln^2 xdx &= \int \ln^2 xd(\frac{1}{2}x^2)\\
&=\frac{x^2}{2}\ln^2 x-\int \frac{1}{2}x^2 2\ln x\frac{1}{x}\\
&=\frac{x^2}{2}\ln^2 x-\int \ln xd(\frac{1}{2}x^2)\\
&=\frac{x^2}{2}\ln^2 x-(\frac{x^2}{2}\ln^2 x-\int \frac{1}{2}x^2\frac{1}{x}dx)\\
&=\frac{x^2}{2}\ln^2 x-\frac{x^2}{2}\ln^2 x+\frac{x^2}{4}+C
\end{aligned}
$$

$$\int \text{幂} * \text{三角} dx$$

$$\text{求} \int x^2\sin 2xdx.$$

$$
\begin{aligned}
\int x^2\sin 2xdx &= -\frac{1}{2}\int x^2d(\cos 2x)\\
&=-\frac{1}{2}(x^2\cos 2x-2\int x\cos 2xdx)\\
&=-\frac{x^2}{2}\cos 2x+\frac{1}{2}\int xd(\sin 2x)\\
&=-\frac{x^2}{2}\cos 2x+\frac{1}{2}(x\sin 2x-\int \sin 2xdx)\\
&=-\frac{x^2}{2}\cos 2x+\frac{x}{2}\sin 2x+\frac{1}{4}\cos 2x+C
\end{aligned}
$$

$$\int x\tan^2 xdx$$

$$
\begin{aligned}
\int x\tan^2 x dx &= \int x(\sec^2 x - 1)dx \\
&= \int x\sec^2 x dx - \frac{x^2}{2} \\
&= \int x d(\tan x) - \frac{x^2}{2} \\
&= x\tan x - \int \tan x dx - \frac{x^2}{2} \\
&= x\tan x + \ln|\cos x| - \frac{x^2}{2} + C
\end{aligned}
$$

$$\int \text{幂} * \text{反三角} dx$$

$$\text{求} \int x\arctan x dx.$$

$$
\begin{aligned}
\int x\arctan x dx &= \int \arctan x d(\frac{x^2}{2}) \\
&= \frac{x^2}{2}\arctan x - \frac{1}{2}\int x^2 \frac{1}{1+x^2} dx \\
&= \frac{x^2}{2}\arctan x - \frac{x}{2} + \frac{1}{2}\arctan x + C
\end{aligned}
$$

$$\int e^{ax} \begin{cases} \cos bx \\ \sin bx \end{cases} dx$$

$$\text{求} \int e^{2x}\cos x dx.$$

$$I = \int e^{2x}\cos x dx = \int e^{2x} d(\sin x) = e^{2x}\sin x - 2\int e^{2x}\sin x dx$$

$$= e^{2x}\sin x + 2\int e^{2x} d(\cos x) = e^{2x}\sin x + 2(e^{2x}\cos x - 2\int e^{2x}\cos x dx)$$

$$= e^{2x}\sin x + 2e^{2x}\cos x - 4I$$

$$\Rightarrow \text{原式} = \frac{1}{5}e^{2x}(\sin x + 2\cos x) + C$$